

Scalability & Future Plan

Date	30 September 2026
Team ID	SWTID-2026-9986
Project Name	AI-ML & GEN AI – Study Assistant (Smart Study Companion)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited to a small number of concurrent users	Slower response time during high usage	High
2	Basic model (limited to few AI features)	Less accurate / less personalized recommendations	Medium
3	No offline access	Cannot be used without internet	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles small number of users	Move to cloud infrastructure (e.g., AWS / Azure); use load balancing and auto-scaling.
2	Data Storage	Local / limited storage	Use scalable database (e.g., Firebase / cloud database); implement data partitioning.
3	Performance	Normal	Optimize code, use caching and efficient model loading; reduce response time.
4	Security	Basic authentication	Implement OAuth / secure login, data encryption and role-based access.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Advanced AI model (more accurate & personalized recommendations)	3 months	Better accuracy and user satisfaction.
Phase 2	Mobile application (iOS & Android)	6 months	Wider accessibility and larger user base.
Phase 3	Offline mode & data sync	9 months	Continuous usage without internet.
Phase 4	Multi-language support	12 months	Global reach and inclusivity.

Phase 5	Integration with learning platforms (e.g., LMS)	15 months	Seamless academic workflow and more features.
Phase 6	Voice assistant & chat support	18 months	Hands-free and interactive user experience.
Phase 7	Advanced analytics & progress tracking	24 months	Better insights for users and institutions.